

MANGAMMAGARI GNANA CHAITHANYA

AI/ML Engineer | Agentic AI, LLM Systems & Deep Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2.5+ years of experience in Generative AI, LLMs, agentic systems, and deep learning. Specialised in end-to-end AI backend development, LLM evaluation pipelines, multi-agent architectures, and cloud deployment across AWS, Azure, and GCP. Proven track record of taking AI prototypes to production for enterprise clients in EdTech, media, automotive, and healthcare — owning accuracy, reliability, and inference cost on live systems.

PROFESSIONAL EXPERIENCE

DataToBiz Pvt. Ltd | AI/ML Engineer

Jan 2026 – Present

(Remote)

- Collaborated directly with enterprise clients across EdTech, media, automotive, and market-intelligence domains to gather requirements, define AI solution architecture, and deliver complete AI backends as the sole AI engineer on each engagement.
- Took a live handwritten-exam grading platform from a fragile single-user prototype to a concurrent-user production service on an 8 GB AWS instance — introducing UUID-based job isolation, disk-based PDF streaming, strict concurrency caps, adaptive DPI, and automated cleanup, which eliminated data corruption, out-of-memory crashes, and server freezes entirely.
- Re-architected the same platform's Gemini 2.5 Pro and Vision pipeline for parallelism using dual-channel rate governance, leaky-bucket bursting, jittered exponential backoff, and model tiering, cutting end-to-end processing by ~48% (31 → 16 minutes per job) with zero accuracy loss.
- Designed and shipped a multi-lane specialist grading router with an LLM archetype classifier and Python-computed mark arithmetic, reducing mean absolute error from 13.4 to ~4 marks and achieving teacher-exact scores across multiple subjects with no regressions.
- Owned LLM inference cost across production pipelines — delivered 37% cost reduction through payload compaction, once-per-paper memo extraction, lane batching, and cover-page skip logic, and evaluated four further cost levers with controlled experiments, closing three with documented evidence.
- Built production-grade agentic AI systems and RAG pipelines using LangChain, LangGraph, and CrewAI, integrating Gemini, GPT-4o, Claude, and Perplexity behind multi-provider abstractions with automatic failover.
- Engineered crash-safe asynchronous backends with FastAPI, Flask, Celery, and Redis — disk-backed execution state, idempotent stage resumption, file-based execution locking, and automatic recovery managers that re-queue interrupted runs on restart.
- Managed cloud infrastructure across AWS (EC2, S3, ECR), Azure (VM, Blob Storage, Network Security Groups), and Hetzner, deploying with Docker, GitLab CI, and systemd service management on Linux.
- Used PostgreSQL for result storage, stage-wise cost tracking, execution status, and structured grounding data across all production pipelines.
- Ran client-facing scrum and technical calls, authored statements of work, root-cause analyses, and cost-optimisation reports, and drove handover and integration specifications for client platform teams.

DataToBiz Pvt. Ltd | AI Engineer Intern

Jul 2025 – Dec 2025

(Remote)

- Built and deployed a 9-stage predictive-maintenance pipeline for medical devices — ingestion, cleaning, feature engineering, SMOTE, BiLSTM and Autoencoder training, SHAP explainability, and FastAPI serving — reducing unplanned failures by 15% and accelerating root-cause diagnosis by 40%.
- Deployed ML services as systemd-managed FastAPI and Streamlit processes on client Azure VMs, resolved the network security group configuration blocking external access, and delivered the client knowledge-transfer session.
- Developed multi-agent GenAI products with CrewAI and LangChain, including a three-agent pitch-deck generator combining SerperDev research, Pinecone RAG retrieval, and GPT-4o-mini content generation with editable .pptx export, cutting preparation time from days to under 30 minutes.
- Built the GPT-4o synthesis layer for a sales-intelligence product, aggregating 12+ external sources into 11-tab company briefs delivered via SSE streaming, owning prompt engineering, retry logic, and structured output validation.
- Production-hardened an n8n lead-sourcing workflow with multi-key API validation and automatic failover, duplicate-record guards, and Gmail error alerting, taking it from demo to reliably live.
- Researched Small Language Models including quantization, LoRA, and QLoRA, converted the AMI Meeting Corpus into per-segment training JSONL with LLM-generated synthetic summaries, and delivered the internal enablement session.

AI SPRY | Data Science Intern

Mar 2024 – Mar 2025

(Hyderabad, Telangana, India)

- Improved dermatology triage accuracy by 35% using YOLOv8 for skin-lesion detection, and improved IVF embryo grading accuracy by 25% with a custom CNN classifier.
- Built an LLM clause-extraction pipeline that cut contract review time by 40%, and trained an LSTM coal-demand forecasting model delivering ~Rs. 1L in annual savings via real-time Streamlit dashboards.

- Benchmarked DeepFace, FER, and RMN models on automotive driver-monitoring video datasets, producing per-frame emotion CSVs, confidence scores, and annotated output videos for engineering evaluation.

CERTIFICATION

- AI & Deep Learning | 360DigiTMG
- Data Science | 360DigiTMG
- Python for Data Science | 360DigiTMG
- SQL for Data Analysis | 360DigiTMG
- Power BI | 360DigiTMG

KEY PROJECTS

AI Exam Grading & Annotation Platform

Dec 2025 – Present

- Multi-lane specialist grader router driven by an LLM archetype classifier, with all mark arithmetic computed in Python; MAE reduced 13.4 → ~4 across five subjects
- Vision-coordinate annotation engine placing tick marks, per-question marks, and student report blocks on scanned scripts, with diagram-aware placement and flipped-page correction
- Validation and reannotation APIs with configurable strictness levels and low-confidence flagging for human review
- Built for scale: 107 papers / 4,754 questions audited; 16 students across 78 pages graded in 42 minutes

Synthetic Persona & Brand Visibility Engine

Jan 2026 – Mar 2026

- Six-stage FastAPI and Celery pipeline generating brand personas from audience research data, crash-safe with idempotent stage resumption and automatic recovery
- Deterministic Share-of-Voice and competitive gap analysis using stable hashed query identifiers
- OpenAI Batch API integration delivering 50% cost savings on the simulation stage; fully data-agnostic across any country, category, or source schema

Rider Emotion Recognition Pipeline

Feb 2026 – Apr 2026

- Benchmarked DeepFace, RMN, HSEmotion, Vision Transformer, YOLO11-L, and MediaPipe on driver-monitoring video; established a 72.09% baseline
- Disproved two candidate approaches through controlled ablations, tracing degraded accuracy to shared training-set bias rather than tuning
- One-second majority-vote label stabilisation, frontal and low-light frame filtering, and eyewear handling; runtime cut from 20 to 13 minutes

Conversational Voice + Chat Agent

Dec 2025

- Retell AI voice agent with sub-200 ms WebRTC latency, knowledge-base grounding, and guardrails preventing pricing or timeline speculation
- Post-call lead capture and live calendar slot-checking with appointment booking; deployed on Hetzner cloud with a client-facing cost model

PERSONAL PROJECTS

NetworkSecurity — Phishing Detection with a Full MLOps Stack | [GitHub](#) | Python · MLflow · Docker · GitHub Actions · AWS ECR / EC2 · FastAPI

- Operationalised phishing-URL detection as a full ML system rather than a notebook: modular training pipeline with MLflow tracking and model registry, containerised inference, and GitHub Actions auto-pushing to AWS ECR on every merge, with the FastAPI endpoint deployed on EC2.

AdvRAG — Hybrid Vector + Keyword Retrieval System | [GitHub](#) | Python · LangChain · FAISS · ChromaDB · BM25 · OpenAI / Gemini Embeddings

- Built a hybrid retriever fusing FAISS semantic search with BM25 keyword scoring via reciprocal rank fusion to fix vector-only RAG failure on keyword-dense queries; compared recursive, semantic, and fixed chunking across Hit Rate and MRR with documented latency-accuracy trade-offs.

TravelAgent — Graph-Based Multi-Agent Travel Planner | [GitHub](#) | Python · Gemini Pro · LangGraph · LangChain · Flask

- Designed a LangGraph directed state graph decomposing open-ended travel requests into Research, Budgeting, Routing, and Reporting agents over shared state, choosing graph orchestration over a linear chain so budget constraints can conditionally re-route the research agent.

TECHNICAL SKILLS

- **Programming Languages:** Python (advanced), SQL, JavaScript / Node.js, HTML, CSS
- **LLMs & Generative AI:** Gemini 2.5 Pro / Flash / Vision, GPT-4 / 4o / 4o-mini, Claude (Opus, Sonnet), LLaMA 3, Mistral, Perplexity API, RAG, Prompt Engineering, Structured Output (Pydantic / Instructor), QLoRA, LoRA, Quantization, Small Language Models, LLM Cost Optimisation, Prompt Caching, Batch APIs

- **Agent Frameworks & Orchestration:** LangChain, LangGraph, CrewAI, AutoGen, Multi-Agent Systems, Supervisor / Router Architectures, Tool Calling, Model Context Protocol (MCP), n8n
- **Backend & Distributed Systems:** FastAPI, Flask, Celery, Redis, Async Pipelines, SSE Streaming, WebRTC, Retell AI, REST API Design, Job Isolation & Idempotent Resumption, Rate Limiting & Backoff
- **Deep Learning & Computer Vision:** TensorFlow, PyTorch, Hugging Face Transformers, CNN, BiLSTM, LSTM, Autoencoders, Vision Transformers, YOLOv8 / YOLO11, DeepFace, MediaPipe, OpenCV, SHAP, LIME, SMOTE, Transfer Learning, Fine-Tuning
- **Machine Learning:** Classification, Regression, Clustering, Time-Series Forecasting, Anomaly Detection, Predictive Maintenance, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Evaluation
- **Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, SQLite, Pinecone, ChromaDB, FAISS, Milvus, Qdrant
- **Cloud Platforms & MLOps:** AWS (EC2, S3, ECR, IAM), Azure (VM, Blob Storage, Network Security Groups), GCP (Document AI, Vision AI), Hetzner, Render, Docker, MLflow, Airflow, GitHub Actions CI/CD, GitLab CI, Model Versioning & Monitoring
- **Development & Delivery:** Git, GitHub, GitLab, Streamlit, Postman, Linux / systemd, Power BI, Agile & Scrum, Client-Facing Delivery, Technical Documentation & SoW Authoring

EDUCATION

Siddharth Institute of Engineering and Technology

2020 – 2024

B.Tech in Computer Science Engineering (AI & ML Specialisation) | CGPA 8.17 / 10

Relevant coursework: Deep Learning, Natural Language Processing, Computer Vision, Machine Learning, Data Structures & Algorithms.